

Description of Work (DOW) for: Network Engineering Services (SMA1)

1.1 Objectives

The Contractor shall provide Network Engineering Services that include designing, implementing, and maintaining secure network architectures that enforce Zero Trust principles, including network segmentation, micro-segmentation, and least privilege access controls. The Contractor will ensure that all network devices including switches, routers, and perimeter systems are configured, hardened, and continuously monitored in accordance with federal security standards and industry's best practices. The Contractor shall provide engineering and operational support for the Congressional Budget Office's Cisco network environment. The objective of this support is to strengthen the security, reliability, and resilience of the network infrastructure while addressing identified Incident Response (IR) remediation requirements.

1.2 Scope of Work

The Contractor shall provide engineering and operational support to enhance, secure, and sustain CBO's enterprise network infrastructure. This effort supports the remediation of identified Incident Response (IR) findings, strengthens overall network security posture, and ensures alignment with federal cybersecurity standards, including NIST SP 800-53, NIST SP 800-207 (Zero Trust Architecture), and Cisco best practices. The Contractor will assist in the design, implementation, and maintenance of secure network architectures across CBO's Cisco-based environment, including core, distribution, access, and edge layers, with a focus on enforcing segmentation, least privilege access, and secure network access controls such as 802.1X.

The scope of work includes, but is not limited to, the following activities:

- **Enterprise Network Operations:** Support the implementation, operation, and optimization of the Congressional Budget Office's Cisco-based network infrastructure, including core, distribution, access, and edge environments, ensuring reliability, performance, and availability in support of mission requirements.
- **Network Security & Compliance Support:** Assist in implementing and maintaining network security controls aligned with NIST SP 800-53, NIST SP 800-207 (Zero Trust Architecture), and Cisco security best practices, including participation in monitoring, risk mitigation, and maintaining network security posture.
- **Routing, Switching & Infrastructure Services:** Configure, manage, and troubleshoot routing, switching, and network services (e.g., VLANs, DNS, DHCP, VPNs), ensuring secure configurations and alignment with organizational standards.
- **Network Monitoring & Operations Support:** Assist in monitoring network performance and availability, responding to alerts, and supporting integration with enterprise monitoring, logging, and security tools to enhance visibility and incident response.
- **Vulnerability Management & Device Maintenance:** Support network device patching, firmware updates, and vulnerability remediation activities, ensuring infrastructure components remain secure, compliant, and up to date.

- **Change Management & Documentation:** Participate in changing management processes, maintain accurate network diagrams and configuration documentation, and support the development and upkeep of standard operating procedures to ensure operational continuity and audit readiness.
- **Customer Service & Team Collaboration:** Serve as the technical adviser for complicated service desk tickets and modifications to better support network operations, while collaborating with cloud, Microsoft engineering, and cybersecurity teams.

1.2.1 Specific requirements / Tasks

- Implement and maintain network security controls aligned with NIST SP 800-53, including access control (AC), configuration management (CM), system and communications protection (SC), and audit and accountability (AU) control families.
- Enforce Zero Trust network architecture principles in accordance with NIST SP 800-207, including network segmentation, micro-segmentation, and continuous verification of users and devices.
- Design and implement least-privilege network access controls, ensuring role-based and identity-aware access across all network layers.
- Manage 802.1X port-based network access control to prevent unauthorized device connectivity and enforce authentication at the network edge.
- Configure and maintain centralized logging and audit capabilities for all network devices, ensuring logs are forwarded to enterprise SIEM platforms and retained in accordance with compliance requirements.
- Conduct continuous monitoring and vulnerability assessments of network infrastructure, identifying risks and coordinating remediation in alignment with NIST Risk Management Framework (RMF) practices.
- Harden all network devices using secure configuration baselines (e.g., Cisco Secure Configuration Guides), including disabling unnecessary services, enforcing strong encryption protocols, and securing management interfaces.
- Secure public-facing and perimeter network assets by implementing strict ingress/egress filtering, firewall rule optimization, and multi-factor authentication for administrative access.
- Support incident response activities by providing network-level analysis, containment actions (e.g., segmentation, blocking malicious traffic), and forensic data collection.
- Establish and maintain secure network segmentation strategies to limit lateral movement and protect high-value assets and sensitive environments.
- Ensure all network changes follow formal change control processes with security impact analysis, supporting compliance with NIST configuration management requirements.
- Develop, implement, and maintain Network Standard Operating Procedures (SOPs); review and update all SOPs on at least an annual basis or as required to reflect changes in technology, policy, or security requirements.

- Document and maintain detailed hardware and configuration baselines for all network devices, including Cisco switches, routers, firewalls, and related infrastructure; conduct annual reviews and updates.
- Perform root cause analysis (RCA) for network incidents, including performance degradation, outages, and security events; document findings and implement corrective and preventive actions.
- Maintain automated network patch management and firmware update procedures in accordance with Cisco best practices and organizational security policies.
- Develop, maintain, and update comprehensive network diagrams that accurately reflect the CBO enterprise network architecture, including cloud, production, and secure environments; review and update diagrams annually or as changes occur.
- Troubleshoot and maintain DNS services, including configuration changes, issue resolution, and performance optimization.
- Support continuous, real-time monitoring of network infrastructure (24/7 operations), including integration with network management and security monitoring tools.
- Maintain accurate and up-to-date documentation of network configurations, assets, and operational procedures to support audit readiness and operational continuity.

1.3 Key Personnel

The following role is anticipated to be required for this work and is considered Key Personnel, in accordance with SOW paragraph 2.9.

- Network Engineer

1.3.1 Key Personnel Requirements

The Contractor shall provide personnel who meet the following specific requirements:

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1.3.2 Required Credentials

The candidate(s) provided by the Contractor must have the following current credentials:

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These (or their equivalent prior certifications) should have been maintained for a minimum of 5 years. Expired certifications or certifications never used professionally will not be considered.

1.4 Government Furnished Equipment and Information

Pursuant to SOW paragraph 2.5, Government-Furnished Property, the Government will provide the following to perform the tasks outlined in this Description of Work:

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